The First Distinction

A Companion to Void and Form

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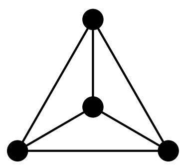

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Machine-verified in Agda

June 2026

Prologue: The Old Dream

A simple observation stands behind this book. Strong theories often explain more because they work with fewer independent assumptions. Not a separate ultimate explanation for every domain, but as few pieces as possible from which more follows.

That is what the old dream means here. It does not yet mean a specific physical theory. It means the hope that what appears separate may be brought back to a deeper connection.

Leibniz stands at the beginning of this opening because he states that movement in especially clear form. At first the issue is not planets, light, or spacetime. It is the demand that questions be framed so that one can see what follows from which assumptions. Not opinion against opinion, but inspectable consequence.

In physics the same movement later takes concrete form. Newton shows that the fall of a stone and the motion of the planets do not belong to separate worlds. Maxwell brings electricity, magnetism, and light under one description. Einstein binds space and time into a single structure.

Something comparable happens in mathematics as well. Very different objects are brought under common structures. Numbers, spaces, functions, or symmetries do not merely sit side by side; they become intelligible as cases of more general concepts.

And the same direction appears in logic. It is not enough to collect individual true statements. One asks which few forms of inference carry many particular cases. Here too the emphasis shifts from isolated result to the structure that makes many results intelligible at once.

These examples do not all say the same thing. But they show the same form of progress: what at first looks separate becomes readable as part of a connection. That is why Leibniz, Newton, Maxwell, and Einstein stand here in sequence.

This book belongs inside that movement, but starts earlier. It does not begin by asking which equation might join gravitation and quantum theory, or which model might absorb all particles and forces into one more comprehensive system. It asks something prior.

Is there a common origin still beneath logic, mathematics, and physical description themselves?

That question is larger than a technical unification problem and smaller than a proclamation about reality as a whole. Nothing here claims that the world has already been derived. The issue is whether the possibility of description may rest on a common minimal structure.

This question does not stand in empty space. In different forms it has been prepared already: in Spencer Brown through the drawn distinction, in Bateson through the link between difference and information, in Shannon through the formal treatment of information, in Wheeler through the question of how physical reality may be bound to elementary distinctions. The same tendency returns in relational directions of mathematics and logic.

These names are not offered here as authorities. They indicate only the direction to which this book belongs. A more exact placement can come later. For the opening, it is enough to say: the thought does not begin from zero.

The guiding assumption of the present project can therefore be stated plainly: stable information presupposes a successfully carried distinction. If that assumption fails, the route ends at once. If it holds, the question becomes how far the first distinction can carry.

To examine that assumption and to ask whether it holds, this book uses type theory. It is not the only possible language. But it has one decisive advantage here: it makes the cost of each step visible. Every claim must be paid for by construction, and hidden assumptions cannot pass silently but must show themselves. If such assumptions exist, they do not enter as an intended back door. That much should already be said here.

This companion takes only the first formally controlled step in that longer route. It presents neither the full theory, nor finished physics, nor an accomplished unification. It seeks the first point at which the conditions of stable distinction can be written completely, constructed explicitly, and checked by machine in intensional Martin-Löf type theory with Agda.

Its path is therefore narrow and deliberate. Why distinction at all? What conditions must a stable distinction satisfy? In what formal language can those conditions be made exact? What is the first fully writable carrier of those conditions? What already follows from so little?

Everything beyond that belongs to the larger investigation of *Void* and *Form*. *Void* asks next how much mathematics can be reconstructed under the same formal constraints. *Form* is the later question of whether some of those structures can be read physically and tested against the world. This book marks only the entry into that chain.

Contents

Chapter 1

Why Distinction?

> "Why is there something rather than nothing? For nothing is simpler and easier than something."
>
> Gottfried Wilhelm Leibniz

This chapter shifts the question, as announced, below the level of objects, laws, and theories. Origin, determination, physics, and information are not presented as proofs from four disciplines. They clarify only why the guiding assumption is plausible: determination presupposes successful distinction. Physics has no priority here. It appears only as the place where a boundary of description can be seen especially sharply.

The Question of Origin

In 1714 Leibniz formulated the question of origin in a form that remains sharp: why is there something rather than nothing? Nothing, Leibniz says, is simpler and easier than something. If the world could just as well have been empty, then precisely its not being empty calls for an account.

No measurement answers this question. One can trace the history of the cosmos back to the first physically describable state and still leave open why there is a cosmos whose history can be traced at all. The question reaches beyond everything data alone can decide.

But the question already contains one condition. To ask between *something* and *nothing*, both must be distinguishable. A nothing that could not be distinguished from something could not appear as nothing. The question of origin therefore already presupposes a distinction it does not itself explain.

That is why it leads immediately further. Not only: why is there something rather than nothing? But also: what must already hold for something to be distinguishable from nothing at all?

Determination

A brief answer to that last question is: it must be possible at all to set something off from something else. This condition concerns not only the contrast between something and nothing. It holds wherever something is determined as this and not that.

A simple case is enough.

A mark is this mark and not the blank place on the page. This sheet is this sheet and not that one. One side is this side and not the other. Wherever something is determinate, something else is excluded.

One cannot step behind holding-apart itself in search of a purer ground. Without this elementary difference there would be no determinate something. A sign that is not distinguished from what it refers to is no sign. A mark that does not stand out from the sheet on which it stands is no mark.

At first this describes only the simplest case. Its reach is limited but important: wherever something appears as this and not as that, something must already be distinguished from something else.

The next section takes physics as an especially sharp example. The point is not a physical derivation. It is the prior fact that physics already works with distinguishable states.

The Physical Limit

Modern cosmology reconstructs the history of the universe backward. Galaxies move closer together, the universe becomes hotter and denser, atoms dissolve into plasma, nuclei break apart. This reconstruction is quantitative, tested, and in important regions very precise. Yet it does not reach its first state. The closer physics comes to the beginning, the thinner the data become. Light does not reach us from before a certain point. Energies climb beyond everything an experiment can reproduce. Space and time themselves lose the status of magnitudes that could simply be carried further within the same theory.

One can take this for a mere lack of data. Perhaps, so the natural expectation goes, deeper measurements or a deeper theory will suffice. But that expectation already assumes that better data will in the end also give better answers about origin.

The opening has already shown that strong theories unify much and still work with presuppositions they do not produce from themselves. Here only one point matters for the present question: even the strongest physical description already presupposes distinguishable states and fixed basic magnitudes.

The point can be put more sharply. Physics does not describe the world as an undifferentiated something, but as states, relations, magnitudes, and changes. Already for that it presupposes a field in which differences hold: between this state and that one, this value and another, this relation and another. Remove that distinguishability, and one loses not only data, but the object of physical description itself.

The beginning of the universe is therefore not simply one more state within an already differentiated world. It concerns the question of how a field of distinguishable states can be present at all.

The early edge of physical description is therefore not only an open measurement problem. It shows that physics already reckons with distinguishable states. Remove that presupposition, and no measurement, no dynamics, no information, and no physics as physics remain.

The question is therefore not only: which state was the first? But: what must hold for states to be in force at all? Then the issue is no longer only the universe, but the condition under which anything can appear as different at all. If states occur only as distinguished states, then information too depends on distinction. That raises the next question: why exactly?

Information

Information presupposes distinction. A measured value is this value only because it is not another. A recorded bit is this bit only because a different outcome remains excluded.

Spencer Brown places at the beginning of his calculus the drawn distinction. Wheeler reads physical reality as an answer to yes-or-no questions. In both cases information hangs on retained difference.

For the rest of the inquiry, that becomes a working assumption. Every piece of information rests on a successful act of distinction. *Successful* means: the difference must be stateable, it must not collapse at once, and it must be determinate enough to present itself again as the same.

The further question is therefore not only where a description reaches its limit. It is: what must a distinction do if it is to carry information? Can a difference be stated at all? Does it hold? And is a field of such differences complete, with neither less nor more?

Chapter 2

Conditions of Stable Distinction

The previous question asks after a performance, not a boundary. The answer develops in three stages: a difference must be stateable, it must hold, and an entire field of such differences must stand exactly. No one stage is enough by itself. Only together do they yield the conditions under which a distinction can persist stably.

Distinguishability

What is the minimum without which a distinction does not succeed? Distinction is not a thing here but a performance: the holding-apart of one from another and the maintenance of that difference.

A mark on a page counts as a sign only because it stands out from the page — because something is read *as* mark and the rest *as* background. A measurement counts as a value only because it could have come out differently: this value and not that one. In both cases the issue is not yet counting or structure. It is the elementary possibility that something can appear as this rather than that.

That is the first demand. Before anything can be counted, ordered, or measured, a difference must be available. Call that distinguishability.

Nothing has yet been said about number, order, or stability. Distinguishability is only the availability of a contrast. But it does not suffice. A stated difference could collapse at once. What is needed so that it does not at once fall back into one?

Non-Collapse

What distinguishability leaves open is whether a difference holds as a difference at all. Something can lie apart for a moment and in the next fall back into one. Then the difference carries nothing. No mark would hold anything. No measured value could rest on it. No state could be asserted against another.

That is why a second demand follows distinguishability. A difference does not suffice merely because it occurs once. It must be secured against collapse. Call this demand Non-Collapse.

Long before there are scales, observers, or instruments, it must be possible for a difference not to vanish at once. Measurement does not create that. Recording does not create it. Both already presuppose it.

The same demand appears in physics as well, for example in the black-hole information problem. The issue there is not first of matter or energy, but of whether differences once achieved can be lost without remainder. This is not a proof of the present argument, but it is a physical example of the same form of question.

The demand can now be sharpened. Non-Collapse means: there must be a genuine separation between two positions, and that separation may not fall back into identity. That is exactly why the formal part will later have to work with inequality and equality.

That raises the next question. Is a single difference secured against collapse enough, or must an entire field of such differences stand complete?

Exact Plurality

The next question no longer concerns a single difference, but an entire field of differences. What does it take for a system to be fixed to a determinate plurality: that exactly these positions are here, neither fewer nor more?

This demand can be shown by a small act of counting. Consider a few blank sheets on a table and suppose there is a definite number of them. Two errors are possible, and they pull in opposite directions. First, some of the sheets counted as different might not be different at all: every way of pointing to one then points equally well to another. There are not several sheets, but one sheet pointed at more than once. Call this collapse: the many become fewer than claimed. Second, there may be more on the table than the count has captured: perhaps a sheet at the edge that the accounting never ruled out. Call this surplus: the field contains more than the account covers.

These two errors are distinct. Collapse means fewer than claimed. Surplus means more than claimed. Whoever prevents only collapse may still have too much in the field. Whoever excludes only surplus may still identify two positions.

The demand that answers both has exactly two parts. The named positions must really lie apart. Call that *separation*. And every position of the field must be captured by the account. Call that *cover*. Separation prevents collapse. Cover prevents surplus. Only both together turn named positions into an exact plurality.

With separation and cover we have, for the first time, a specification complete enough to be written down exactly.

Ask where these two conditions can first be met completely, and the answer is: on the smallest field on which anything can be held apart at all, namely a field of two positions. With two, and not before, separation and cover can be written down completely and finitely. Two is not the origin of distinctness. Two is the first place at which its general conditions become fully explicit.

If this is the minimal performance stable information requires, how little must be granted to write it down exactly?

Chapter 3

The Threshold of Formalization

At this point the question changes character.

Up to now the discussion has been exploratory. The issue was whether certain conditions seem necessary for a distinction to persist. But necessity stated in ordinary language is still only a proposal.

The next question is stricter.

Can these conditions be written in a form precise enough that every consequence becomes checkable and every hidden assumption becomes visible?

The inquiry therefore moves from philosophical orientation to formal specification.

From here on, every claim must be paid for by construction.

The previous chapter narrowed the question to a small finite specification: separation and cover on a field of two positions. At that point ordinary language reaches its limit. What is now missing is not another argument, but a language in which every step can be stated exactly and checked.

There is such a language. The one used here is Martin-Löf type theory in the form realized by the proof assistant Agda.

What is granted can be said briefly: a carrier with two positions, a witness that the two are not equal, and an account that leaves no point of the carrier unaddressed. No numbers, no geometry, no physics, and no classical logic beyond what is constructed explicitly.

Everything else must be built.

In this setting, to assert is to construct. A statement is a type, a proof is a term inhabiting that type, and a machine decides whether the term truly has the type claimed. A proof that does not check is simply a file that does not compile.

The argument is formulated in type theory not because it is the only possible carrier, but because it makes the cost of every step visible.

Two settings, fixed at the head of the file, govern everything that follows.

The flag `--safe` closes every escape hatch: no postulates, no principle asserted without proof, no back door through which an unearned result could enter.

The flag `--without-K` withholds a convenient but additional assumption about equality, so that nothing in the development rests on more than the bare identity type.

Under these restrictions, only the things we actually build exist.

If the formalization later seems to do more than this brief list suggests, that is only because much already follows from little. It is not because more was smuggled in beforehand. The order of writing is not the same as the order of the inquiry.

In the inquiry, distinction, Non-Collapse, and exact plurality appeared first. In the file, what appears first is whatever the checker requires in order for those conditions to be sayable at all.

No new content is introduced here. The file merely provides the few means without which separation and cover could not be stated exactly. These constructions belong to the requirements of the formal language, not to the order in which the inquiry itself arrived at its result.

It therefore begins with the bookkeeping required to manage universe levels. Nothing conceptual enters with that step.

```
open import Agda.Primitive using (Level; lzero; lsuc; _⊔_; Setω)
```

That first line is only housekeeping. It gives the checker a way to track sizes consistently, and none of the conceptual weight of the argument rests on it.

The real formal work begins with the thinnest device needed to say when something is the same and when it is not.

That is why equality comes first in the file. Not because sameness would be prior to distinction in the order of the inquiry, but because separation and cover are not even formulable without a minimal way to say same, not the same, and substitution.

The identity type provides exactly that, and at this stage it is not meant to do more.

```
infix 4 _≡_

data _≡_ {ℓ : Level} {A : Set ℓ} (x : A) : A → Set ℓ where
 refl : x ≡ x
```

```
{-# BUILTIN EQUALITY _≡_ #-}

sym : {ℓ : Level} {A : Set ℓ} {x y : A} → x ≡ y → y ≡ x
sym refl = refl

trans : {ℓ : Level} {A : Set ℓ} {x y z : A} → x ≡ y → y ≡ z → x ≡ z
trans refl q = q

cong : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} {B : Set ℓ₂} (f : A → B) {x y : A} → x ≡ y → f x ≡ f y
cong f refl = refl

subst : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} (P : A → Set ℓ₂) {x y : A} → x ≡ y → P x → P y
subst P refl px = px
```

Next the file needs a way to say impossibility. A guard against collapse is worthless unless one can express that an identification is excluded. The empty type gives that means. Negation is then the map into impossibility, and inequality — the formal shape of being kept apart — is the negation of equality.

```
data ⊥ : Set where

⊥-elim : {ℓ : Level} {A : Set ℓ} → ⊥ → A
⊥-elim ()

¬_ : {ℓ : Level} → Set ℓ → Set ℓ
¬ A = A → ⊥

infix 2 _≢_
_≢_ : {ℓ : Level} {A : Set ℓ} → A → A → Set ℓ
x ≢ y = ¬ (x ≡ y)
```

Then the file needs a way to say exactly: one of two, and which one. Cover cannot be expressed by an untagged choice, because that would merge the two cases. The disjoint union keeps the mark, so that a later case analysis can say from which side a witness comes.

```
infixr 2 _⊎_

data _⊎_ {ℓ : Level} (A B : Set ℓ) : Set ℓ where
  inj₁ : A → A ⊎ B
  inj₂ : B → A ⊎ B
```

Finally, the file needs a way to carry a thing together with a statement about it. That is what the dependent pair does, with the ordinary product as its constant special case. In this way an existential claim becomes an actual witness together with a proof about it.

```
infixr 3 _×_
```

```
record _×_ (A B : Set) : Set where
 constructor _,_
 field
  fst : A
  snd : B

open _×_ public

record Σ (A : Set) (B : A → Set) : Set where
 constructor _,_
 field
  fst : A
  snd : B fst

open Σ public
```

With these means in place, the earlier question can for the first time be translated into a single exact object. What does this smallest object look like, the one that carries separation and cover together?

Chapter 4

The First Distinction

The previous question asked how little has to be granted for the minimal conditions of successful distinction to be writable exactly. Here the title of the book is earned. The Distinction Record is the first complete answer. Formally it is a record. In content it is the first fully writable carrier of a stable distinction: not an inventory of a thing, but a certificate of conditions, exactly the data needed for a difference to persist stably as a difference.

```
record Distinction : Set1 where
 field
  S     : Set
  ℓ     : S
  r     : S
  ℓ≠r : ℓ ≠ r
  cover : (x : S) → (x ≡ ℓ) ⊎ (x ≡ r)

open Distinction public
```

Read back into prose, `S` is the carrier, ℓ and `r` are the two positions, ℓ≠`r` is separation, and `cover` is cover. The record contains no more than that. That is its point. It does not primarily describe an object. It describes the minimal configuration under which a distinction can be held fast as a successful distinction. Its first consequence is simple: `cover` says that every point of the carrier reduces to ℓ or `r`.

```
law1-1-cover : (d : Distinction) → (x : S d) → (x ≡ ℓ d) ⊎ (x ≡ r d)
law1-1-cover = cover
```

If every point is one of the two, then it is enough to check a property at ℓ and `r` in order to obtain it for all points. That is the carrier eliminator.

```
Distinction-elim :
 (d : Distinction) →
 {P : S d → Set} →
 P (ℓ d) →
 P (r d) →
 (x : S d) →
 P x
Distinction-elim d {P} pℓ pr x with cover d x
... | inj₁ x≡ℓ = subst P (sym x≡ℓ) pℓ
... | inj₂ x≡r = subst P (sym x≡r) pr
```

Surplus is therefore not only excluded but eliminated by construction. The fourth field does the opposite work. $\ell \neq \mathsf{r}$ forbids the collapse of the two positions into one. Anyone who tries to identify them produces a contradiction the checker rejects.

Taken together, separation and cover yield a stable two. The next question is what self-maps such a carrier admits.

Chapter 5

What Already Follows

Once the first distinction is given, the next question follows at once: what self-maps does such a carrier admit? This chapter has a limited scope. It does not yet deliver the later closure reading of *Void*. It shows only that even at this first formal threshold the space of endomorphisms is fixed to four cases.

Once the carrier is fixed to two classes, self-maps are almost entirely determined by their values on ℓ and `r`. To compare two maps, we use pointwise equality. On this carrier it is especially manageable because `cover` reduces every argument to one of the two boundary cases.

```
infix 4 _≗_
_≗_ : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} {B : Set ℓ₂} → (A → B) → (A → B) → Set (ℓ₁ ⊔ ℓ₂)
_≗_ {A = A} f g = (x : A) → f x ≡ g x

id : {A : Set} → A → A
id x = x
```

Among these maps one is especially important: the map that exchanges the two positions, sending ℓ to `r` and `r` to ℓ. Call it the dual. Applied twice, it returns every point to its starting place.

```
Distinction-dual : (d : Distinction) → S d → S d
Distinction-dual d x with cover d x
... | inj₁ _ = r d
... | inj₂ _ = ℓ d

Distinction-dual-involutive :
 (d : Distinction) →
 (x : S d) →
 Distinction-dual d (Distinction-dual d x) ≡ x
Distinction-dual-involutive d =
 Distinction-elim d proof-ℓ proof-r
 where
```

```
proof-ℓ : Distinction-dual d (Distinction-dual d (ℓ d)) ≡ ℓ d
proof-ℓ with cover d (ℓ d)
... | inj₂ ℓ≡r = ⊥-elim ((ℓ≠r d) ℓ≡r)
... | inj₁ _ with cover d (r d)
... | inj₁ r≡ℓ = ⊥-elim ((ℓ≠r d) (sym r≡ℓ))
... | inj₂ _ = refl

proof-r : Distinction-dual d (Distinction-dual d (r d)) ≡ r d
proof-r with cover d (r d)
... | inj₁ r≡ℓ = ⊥-elim ((ℓ≠r d) (sym r≡ℓ))
... | inj₂ _ with cover d (ℓ d)
... | inj₂ ℓ≡r = ⊥-elim ((ℓ≠r d) ℓ≡r)
... | inj₁ _ = refl
```

The central question is: how many maps are there, counted up to pointwise equality? The boundary table allows exactly four combinations, one for each pair of choices at ℓ and r — the constant-left map, the constant-right map, the identity, and the dual. We name those four cases explicitly.

```
data EndoCase : Set where
 case-constL : EndoCase
 case-constR : EndoCase
 case-id     : EndoCase
 case-dual : EndoCase
```

The four names must now be tied to actual maps. Inside a module parametrized by a fixed distinction, `interpret` turns each label into its endomorphism, and `classify` reads the matching label back from an endomorphism's two boundary values.

```
module K₄ (d : Distinction) where
 Endo : Set
 Endo = S d → S d

 ≗-refl : {f : Endo} → f ≗ f
 ≗-refl x = refl

 ≗-sym : {f g : Endo} → f ≗ g → g ≗ f
 ≗-sym p x = sym (p x)

 ≗-trans : {f g h : Endo} → f ≗ g → g ≗ h → f ≗ h
 ≗-trans p q x = trans (p x) (q x)

 constL : Endo
 constL _ = ℓ d

 constR : Endo
 constR _ = r d

 dual : Endo
 dual = Distinction-dual d
```

```
dual-ℓ : dual (ℓ d) ≡ r d
dual-ℓ with cover d (ℓ d)
... | inj₁ _ = refl
... | inj₂ ℓ≡r = ⊥-elim ((ℓ≠r d) ℓ≡r)

dual-r : dual (r d) ≡ ℓ d
dual-r with cover d (r d)
... | inj₁ r≡ℓ = ⊥-elim ((ℓ≠r d) (sym r≡ℓ))
... | inj₂ _ = refl

interpret : EndoCase → Endo
interpret case-constL = constL
interpret case-constR = constR
interpret case-id     = id
interpret case-dual = dual

classify : Endo → EndoCase
classify f with cover d (f (ℓ d)) | cover d (f (r d))
... | inj₁ _ | inj₁ _ = case-constL
... | inj₂ _ | inj₂ _ = case-constR
... | inj₁ _ | inj₂ _ = case-id
... | inj₂ _ | inj₁ _ = case-dual
```

Reading a function's label and then interpreting it returns the same function: at the two boundary points by direct inspection, and therefore everywhere by the eliminator. The two boundary lemmas this rests on are routine case analyses; they are part of the checked file and are not reprinted here.

```
classify-sound : (f : Endo) → interpret (classify f) ≗ f
classify-sound f x = Distinction-elim d (sound-at-ℓ f) (sound-at-r f) x
```

The label is not merely available; it is forced. No two distinct labels can interpret to the same function — sixteen comparisons, one for each ordered pair of labels, every one discharged by the single separation witness ℓ≠r. That case analysis is routine and lives in the checked file without crowding the page. From it, classification is unique, and reading a label out and back in returns the very same label.

```
classify-unique : (f : Endo) → (c : EndoCase) → interpret c ≗ f → c ≡ classify f
classify-unique f c c≗f =
 interpret-injective c (classify f) (≗-trans c≗f (≗-sym (classify-sound f)))

classify-interpret : (c : EndoCase) → classify (interpret c) ≡ c
classify-interpret c = sym (classify-unique (interpret c) c ≗-refl)
```

Together this says: the four labels are pairwise distinct, and every self-map falls uniquely under one of them.

```
EndoCase-distinct :
 (case-constL ≡ case-constR → ⊥) ×
 (case-constL ≡ case-id     → ⊥) ×
 (case-constL ≡ case-dual   → ⊥) ×
 (case-constR ≡ case-id     → ⊥) ×
 (case-constR ≡ case-dual   → ⊥) ×
 (case-id ≡ case-dual       → ⊥)
EndoCase-distinct =
 (λ ()) ,
 ((λ ()) ,
  ((λ ()) ,
   ((λ ()) ,
    ((λ ()) ,
     (λ ())))))
```

Gathered into two top-level statements, the result reads cleanly: every endomorphism of a distinction is one of the four cases (soundly), and is so in exactly one way (uniquely).

```
k4-classification-sound :
 (d : Distinction) →
 (f : S d → S d) →
 Σ EndoCase (λ c → K4.interpret d c ≗ f)
k4-classification-sound d f = K4.classify d f , K4.classify-sound d f

k4-classification-unique :
 (d : Distinction) →
 (f : S d → S d) →
 (c1 c2 : EndoCase) →
 K4.interpret d c1 ≗ f →
 K4.interpret d c2 ≗ f →
 c1 ≡ c2
k4-classification-unique d f c1 c2 p1 p2 =
 K4.interpret-injective d c1 c2 (K4.≗-trans d p1 (K4.≗-sym d p2))
```

A quick algebraic over-reading is tempting. But what has been shown is only this: the four maps are closed under composition, and they do not form a group because the two constants are not invertible. What is established is the number of cases and their unique classification up to pointwise equality. Any richer algebraic reading — and likewise the later reading of this four-case structure as a K_4 closure — is an additional step.

The next answer has therefore been reached. From a separated and covered binary distinction follows a unique four-case classification of its self-maps. The next question no longer concerns the structure itself, but its status: what has thereby been shown, and what not?

Chapter 6

What Has Been Shown

The status of the result is limited and exact. This book has not proved the world. But it has shown more than a mere precondition. It has shown that a stable distinction, once written exactly under the stated constraints, already carries a nontrivial closed structure.

What has been shown is a small but precise stock: a carrier with two positions, a witness that they are apart, an exhaustion principle with no third case, and exactly four self-maps up to pointwise equality. Under `--safe` and `--without-K`, these statements are theorems in the strict sense: if any one of them failed, the file would not compile.

At this point the levels of the book can be ordered cleanly. Philosophical orientation was the conjecture that origin, determination, and information all point toward distinction. The working assumption was that successful distinction must satisfy minimal conditions. The formal result is the stock the file actually carries. It is on that level that this chapter moves.

What has not been shown is that the world can be derived from nothing, or that every determination necessarily presupposes distinction as a theorem of Agda. Those claims motivate the inquiry; they are not its checked result. Nor does the file show that no other formal carrier could realize the same content. It shows only that this content can be constructed here without additional assumptions.

Precisely because that boundary remains clear, the gain can now be stated more sharply. Once two positions are granted in full view and kept apart, that minimal grant already carries a closed shape. This is the first point at which the conditions of stable distinction become exact and machine-checkable. The step from philosophical question to formal structure remains small. That is exactly why it is robust.

From here one can also state the boundaries cleanly. The compiler judges the formal kernel. Ordinary mathematics judges the derivations built from that kernel. Experiment judges any later reading of the kernel as physics. The separation between those courts is not caution for its own sake. It is a condition of honesty.

The question of origin therefore remains unanswered. But something decisive has shifted. Its first minimal step — the holding-apart of one position from another — is no longer mere intuition. It is present as checked construction, bounded carrier, and four-case self-map structure.

That is still not the whole story. Already at this first threshold one can see that the resulting stock carries more than mere difference.

Chapter 7

What Is Already Visible

Even if this book studies only the first formal threshold, consequences are already visible.

Once two positions lie stably apart, there is not only difference but a finite stock. Its elements are distinguishable and completely accounted for.

Once every point of the carrier reduces to ℓ or r, there is exhaustive case distinction. Properties can be checked at the two boundary points and transferred across the whole carrier.

Once the carrier is fixed, positions can be related and linked by maps. Those maps do not remain indeterminate. They can be compared, classified, and composed.

With identity and duality, symmetry and reversal already appear. With the two constant maps and the two non-constant cases, the space of possible self-maps of the carrier is completely determined.

None of these structures was additionally postulated. They appear as soon as the conditions of stable distinction are made fully explicit. Here one sees for the first time that the distinction record carries more than bare difference.

No arithmetic has yet been reconstructed, no category worked out, and no physics read off. But the direction is visible: finite stock, relation, map, composition, symmetry, and closure become reachable from the same first specification.

And here the decisive open tension also becomes visible. The four-case structure is not only an endpoint. The next question is whether the closure structure that grows out of distinction already carries the conditions of its own emergence. In other words: does it point back to the original distinction, or could one try to put it in distinction's place? This book does not decide that question. It is the threshold to *Void*.

Chapter 8

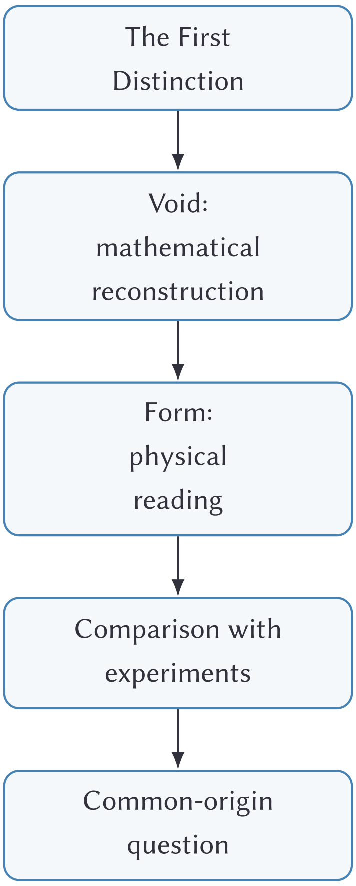
The Reach of Distinction

The opening began with the observation that physics, mathematics, and logic have repeatedly been brought back to deeper common structures. The close of this book can now say more precisely what its own contribution to that movement is. *The First Distinction* shows the first point at which such a movement becomes itself formalizable.

This companion is not the theory as a whole. It is not a physical theory and not a completed unification. But neither is it merely the announcement of a later project. It shows the first load-bearing stone: a formalizable distinction that already carries finite stock, relation, map, composition, symmetry, and a completely determined space of possible self-maps.

From here *Void* pursues the mathematical side of the question. It does not ask which external mathematics may be borrowed, but how much mathematics can be reconstructed under the same discipline. *Form* pursues the other side: whether structures already present in the ledger can be read as structures of physical description and tested against the world.

Above both stands the wider question that motivates the whole project: whether logic, mathematics, and physical description may be different appearances of a common origin.



Void/Form investigates whether logic, mathematics, and physical description can be understood as different appearances of a common structure of distinction. The path to physics is not a theorem of this book, but a later program of interpretation and test. *The First Distinction* supplies the first formally secured step of that investigation.

How far can this go? This book does not answer that question. It only makes the question exact enough to be pursued honestly. The chain may break early, and then its limit will have been identified. Or it may carry far, and then its reach will have been measured rather than proclaimed.

Where a later physical reading is proposed, the formal system does not decide it. There the ledger meets experiment, and science begins in the strict sense.

This book therefore does not end with the bare statement that there are four self-maps. It ends at a sharper threshold: distinction has produced a closed structure, and the next question is whether that structure points back to the conditions of its own emergence. That question is not answered on the first pages of *Void*.

It is prepared across the whole formal reconstruction and decided only at its late end.